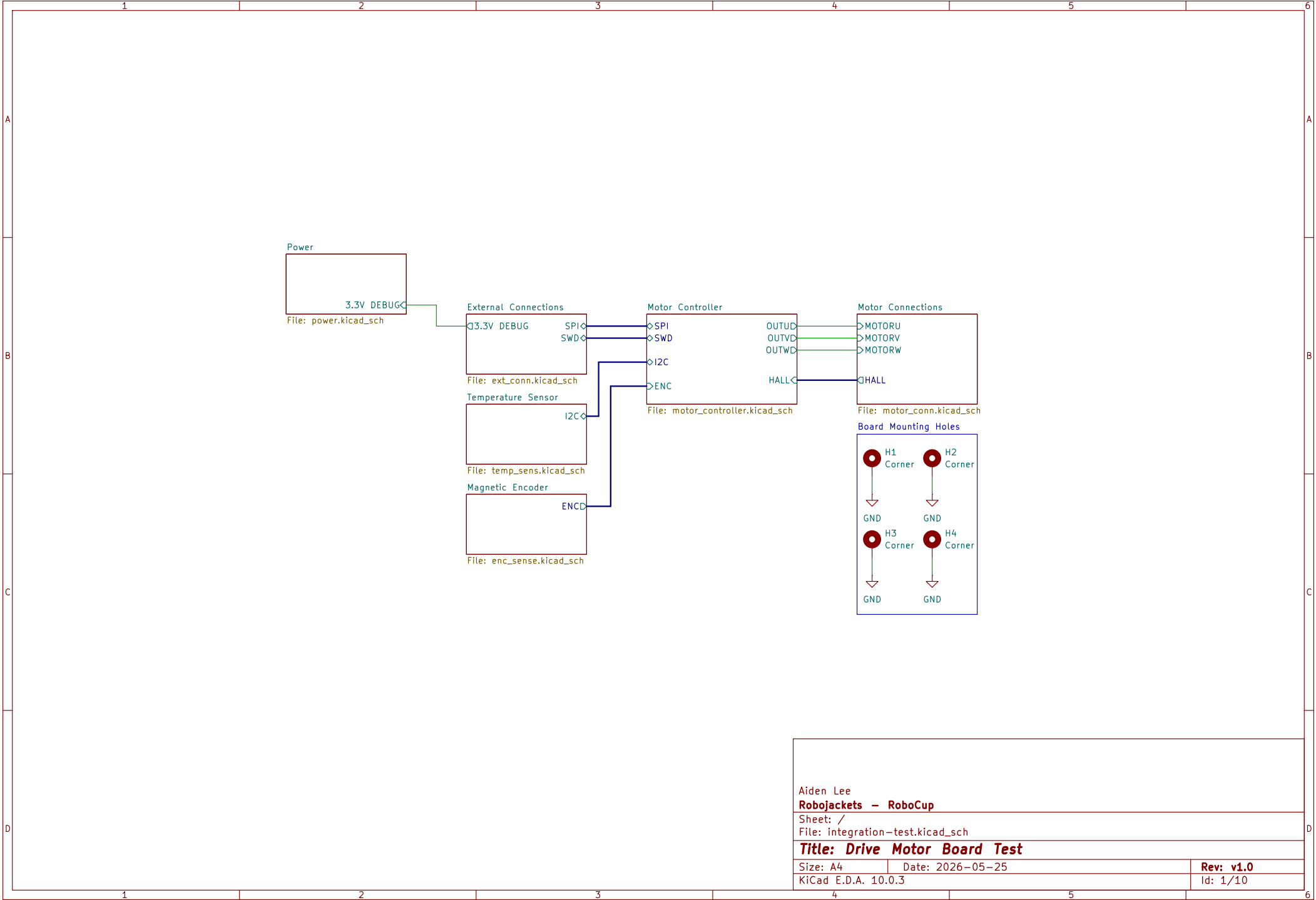




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Sheet: /

File: integration-test.kicad_sch

Title: Drive Motor Board Test

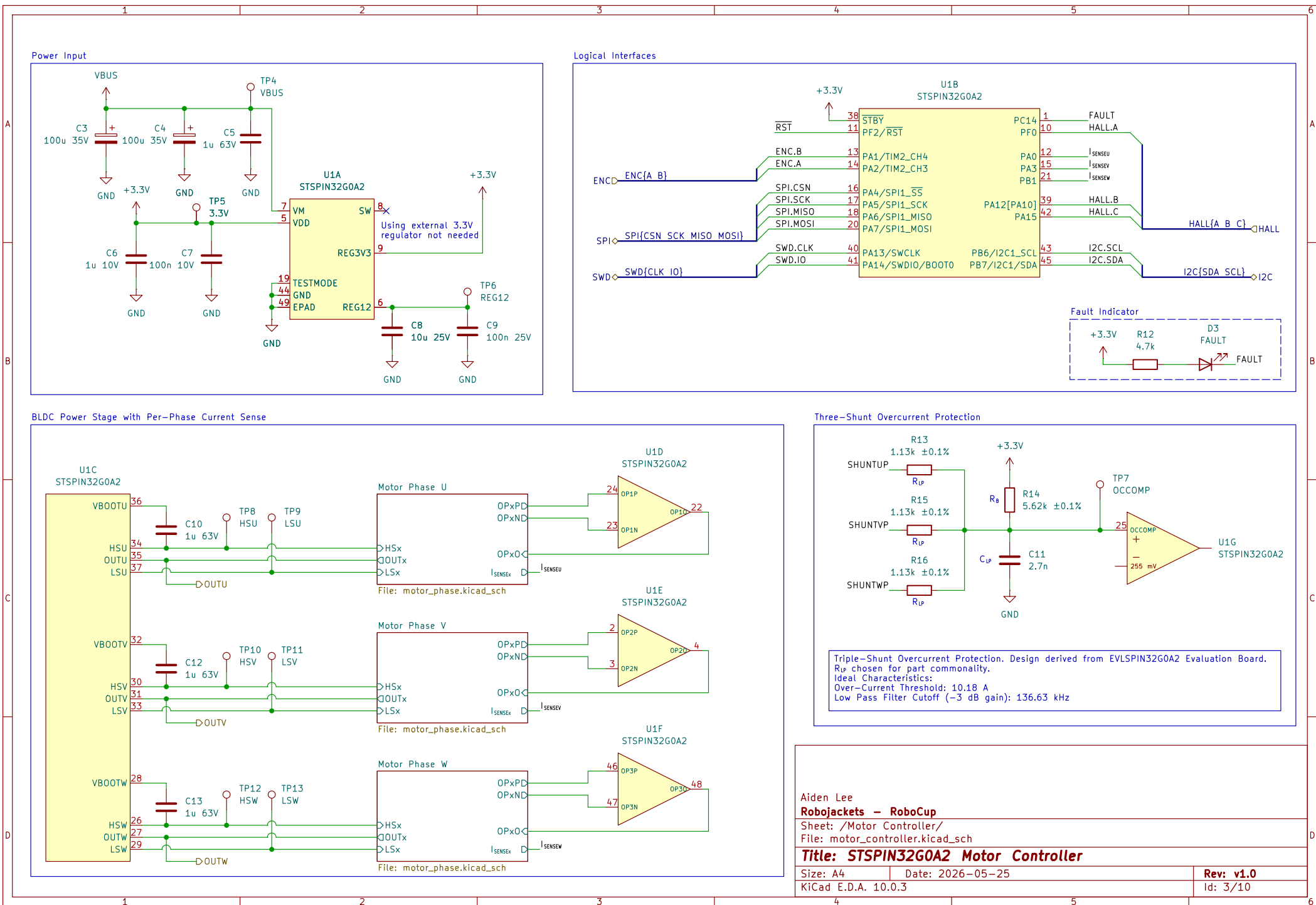
Size: A4

Date: 2026-05-25

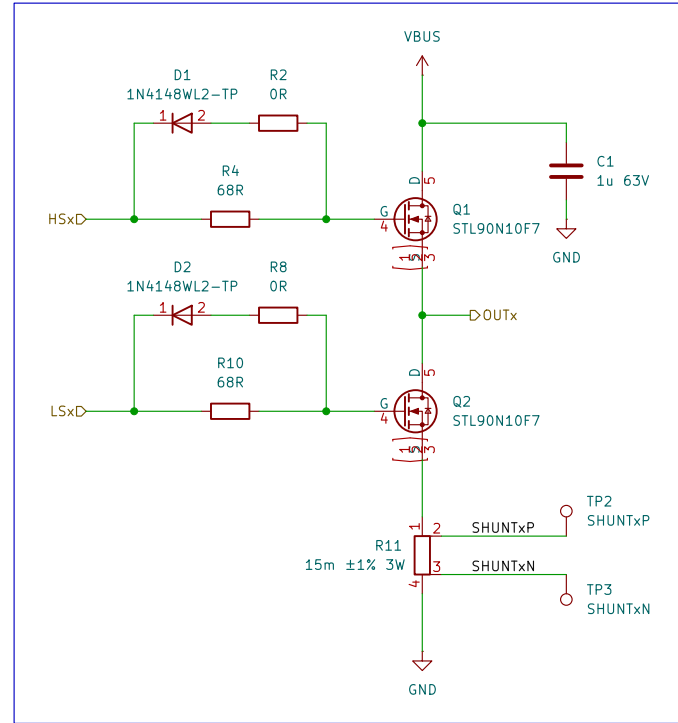
Rev: v1.0

KiCad E.D.A. 10.0.3

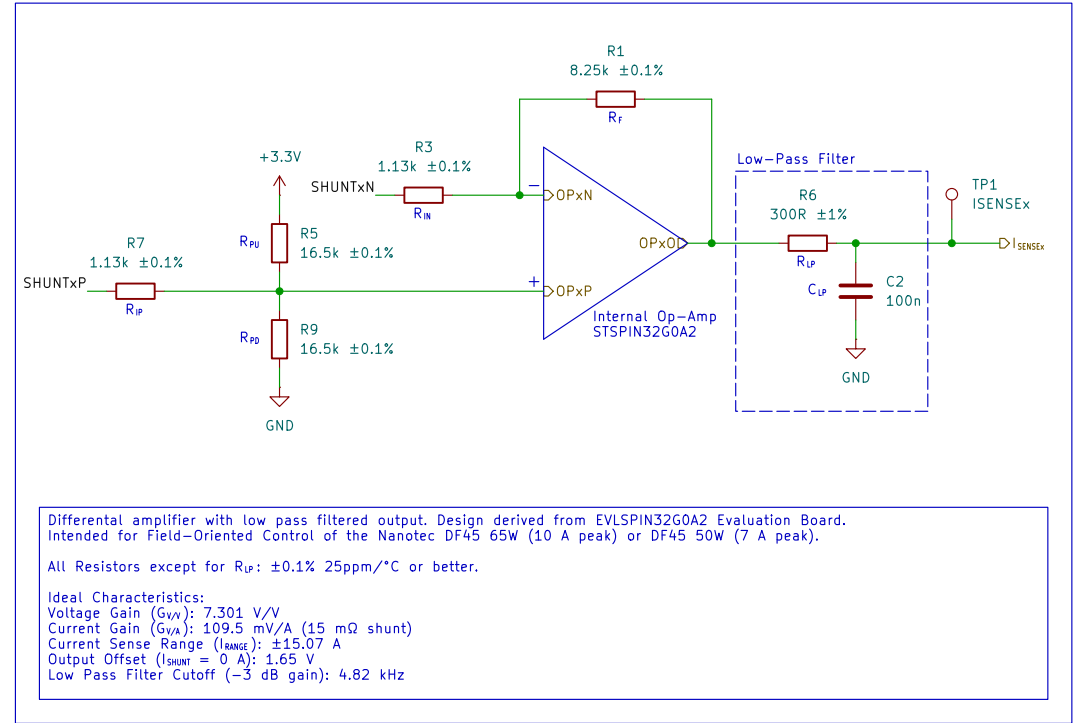
Id: 1/10



Half Bridge for BLDC Motor Phase



Current Sensing Op-Amp with Low Pass Filtered Output



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Sheet: /Motor Controller/Motor Phase U/

File: motor_phase.kicad_sch

Title: Single Motor Phase Power Stage

Size: A4

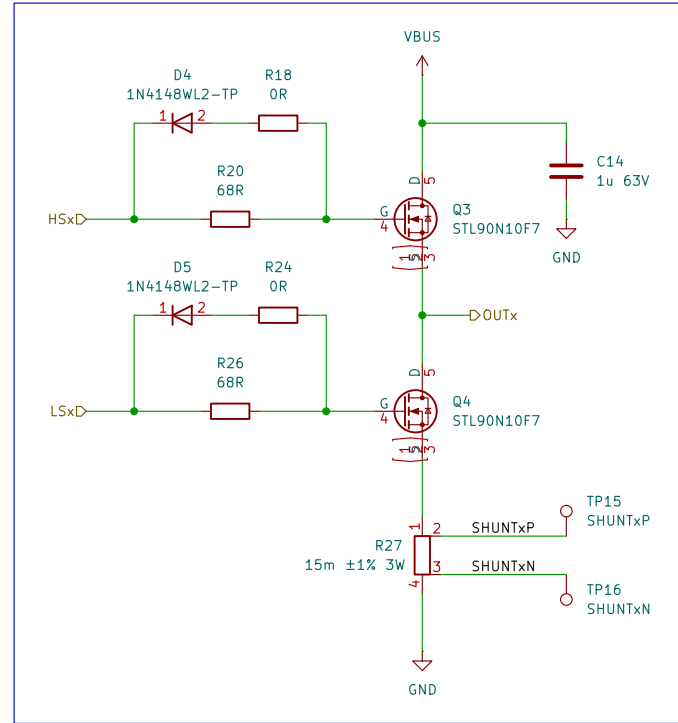
Date: 2026-05-25

Rev: v1.0

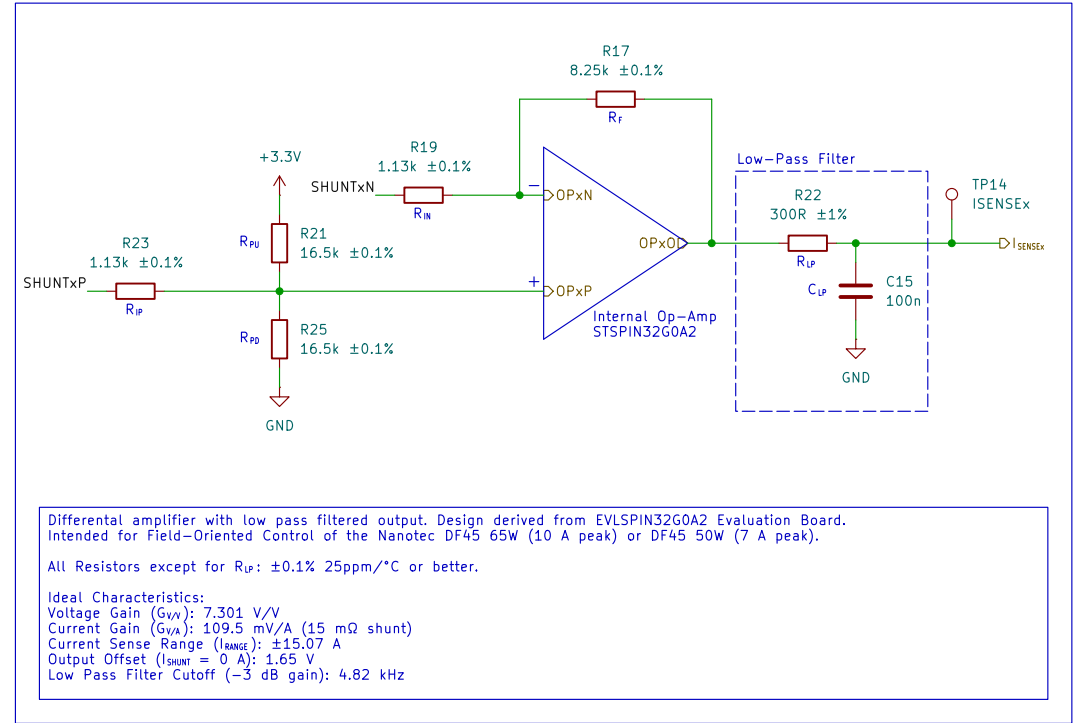
KiCad E.D.A. 10.0.3

Id: 2/10

Half Bridge for BLDC Motor Phase



Current Sensing Op-Amp with Low Pass Filtered Output



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Sheet: /Motor Controller/Motor Phase V/

File: motor_phase.kicad_sch

Title: Single Motor Phase Power Stage

Size: A4

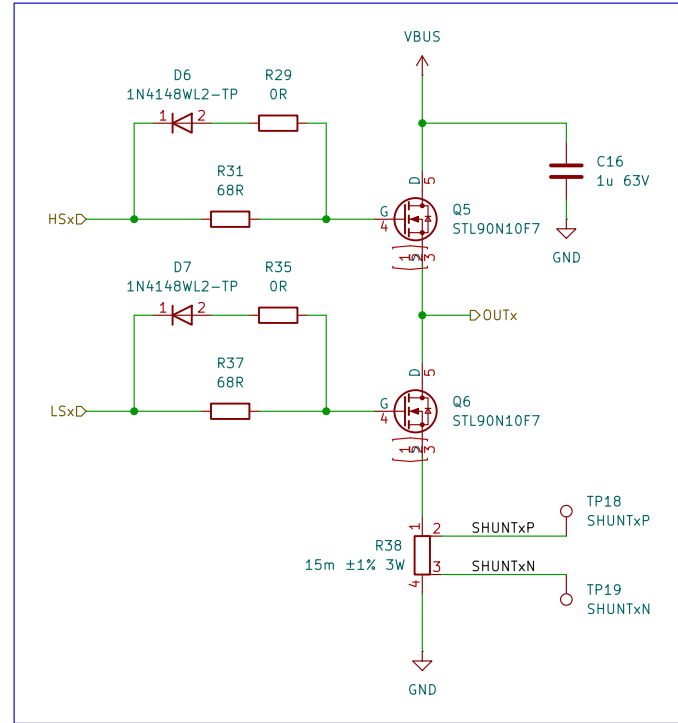
Date: 2026-05-25

Rev: v1.0

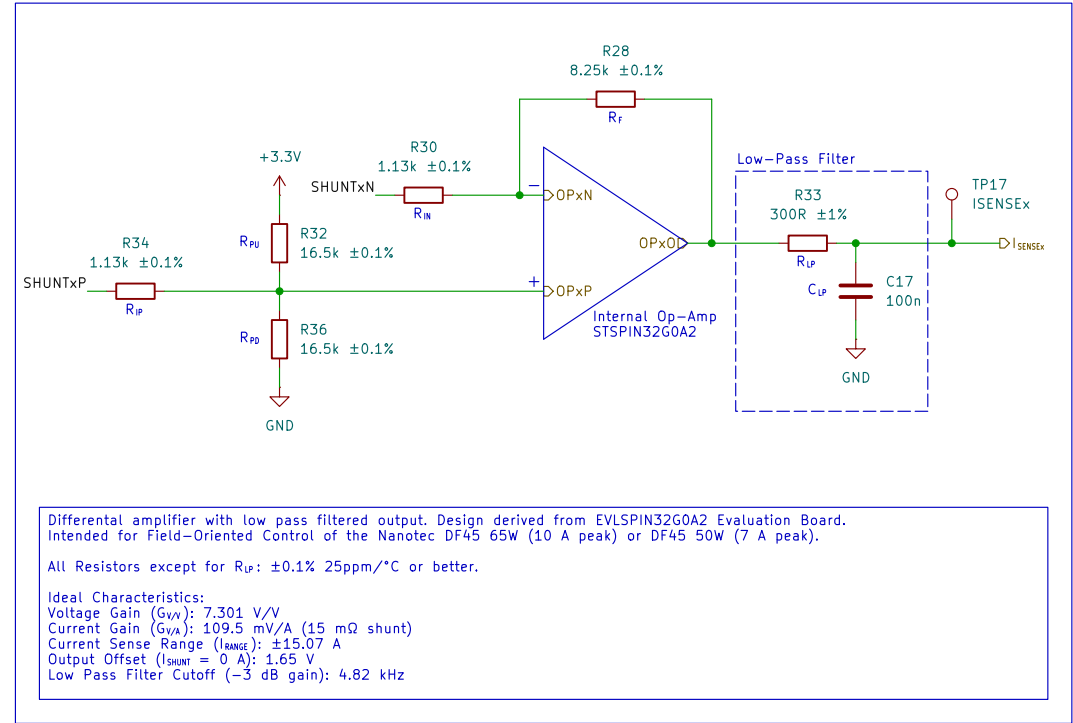
KiCad E.D.A. 10.0.3

Id: 4/10

Half Bridge for BLDC Motor Phase



Current Sensing Op-Amp with Low Pass Filtered Output



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Sheet: /Motor Controller/Motor Phase W/

File: motor_phase.kicad_sch

Title: Single Motor Phase Power Stage

Size: A4

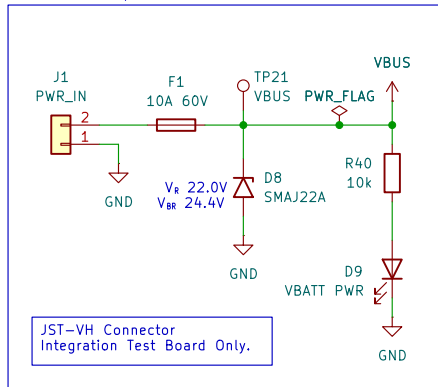
Date: 2026-05-25

Rev: v1.0

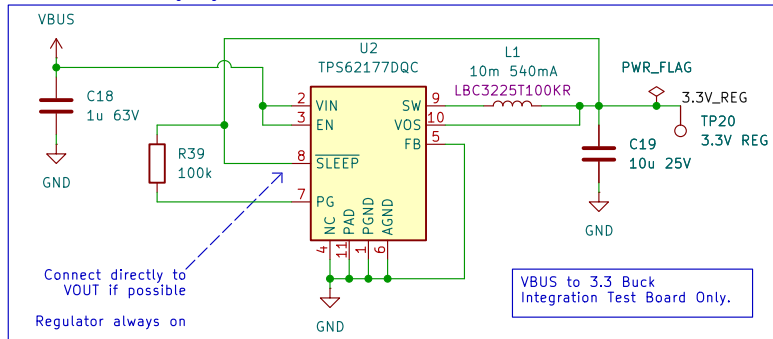
KiCad E.D.A. 10.0.3

Id: 5/10

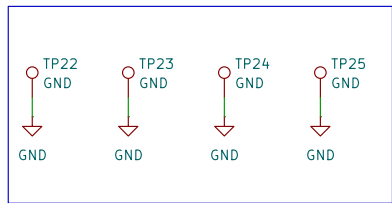
VBATT Power Input



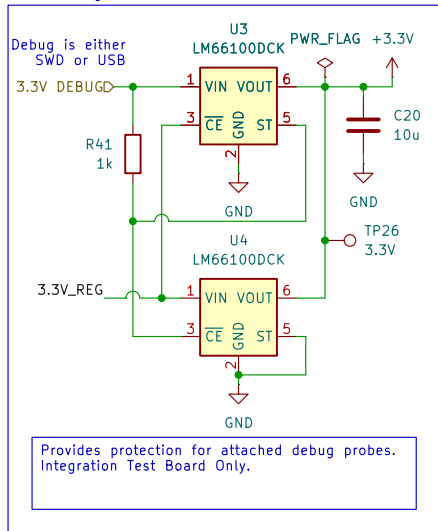
3.3V 500 mA Switching Regulator



Extra GND Test Points



3.3V OR-ing Power Mux



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Sheet: /Power/
File: power.kicad_sch

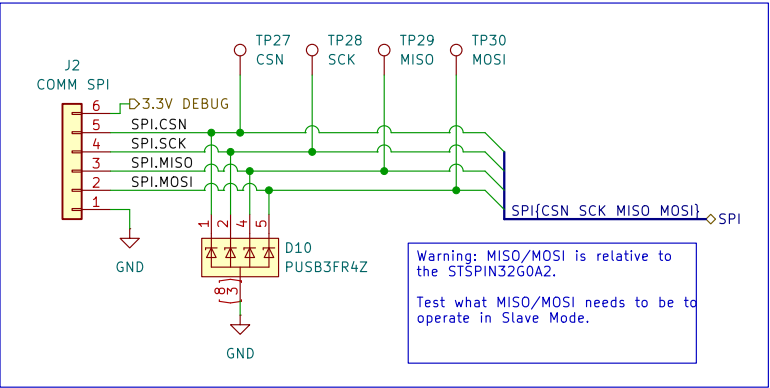
Title:

Size: A4
KiCad E.D.A. 10.0.3

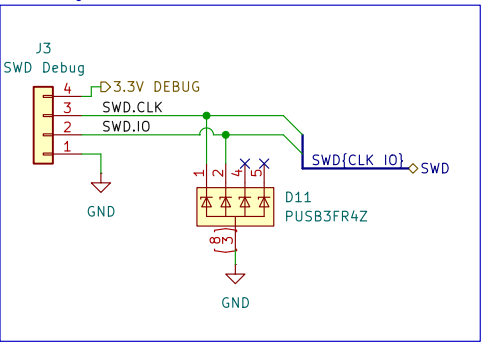
Date: 2026-05-25

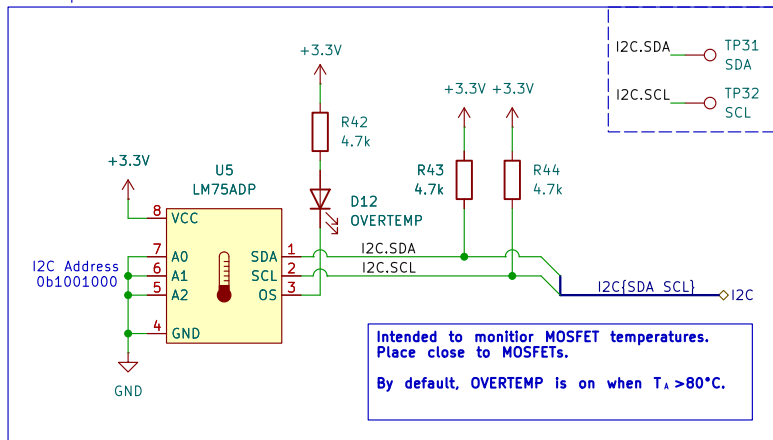
Rev: v1.0
Id: 6/10

External SPI Bus



SWD Debug Header





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Sheet: /Temperature Sensor/

File: temp_sens.kicad_sch

Title:

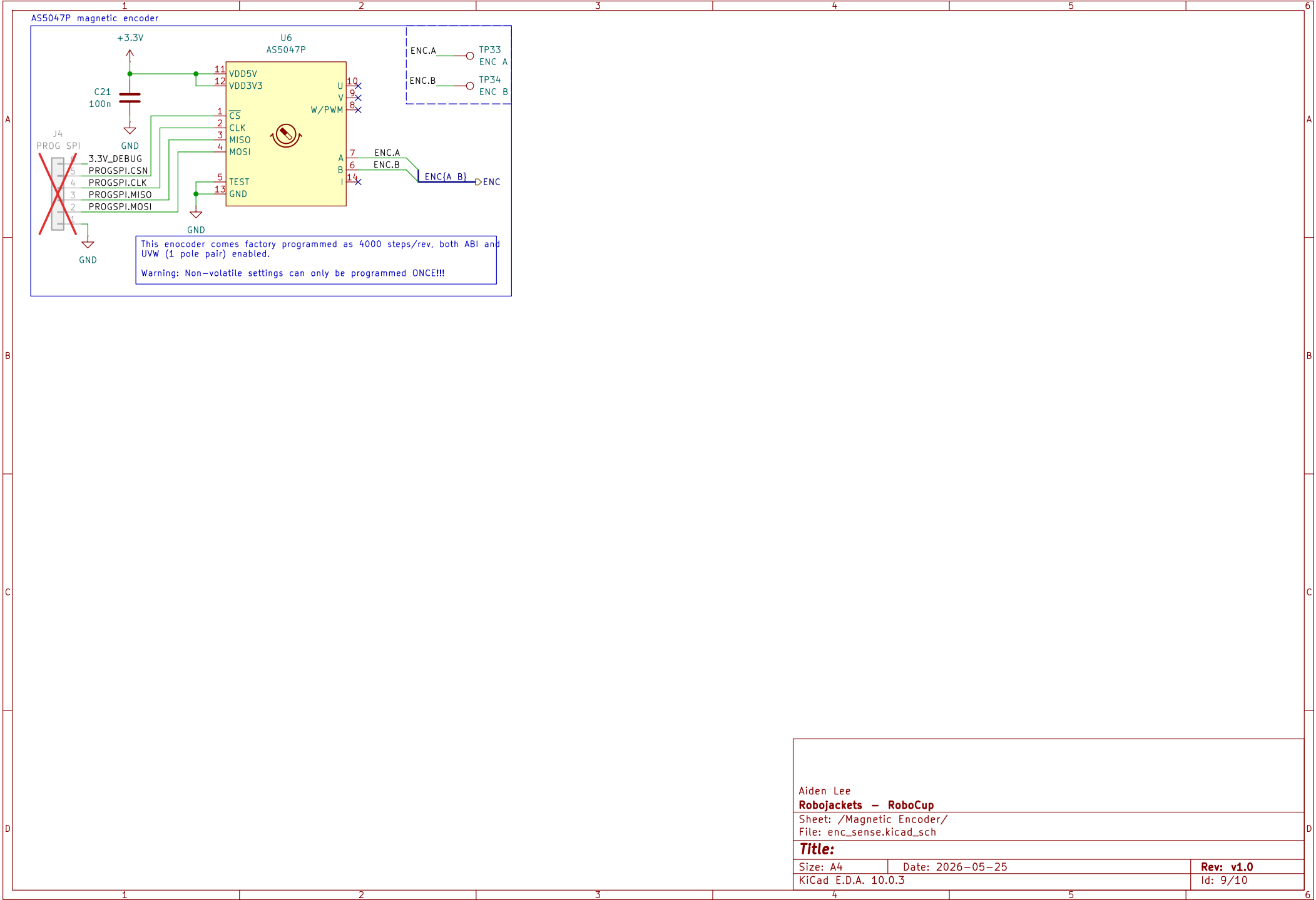
Size: A4

Date: 2026-05-25

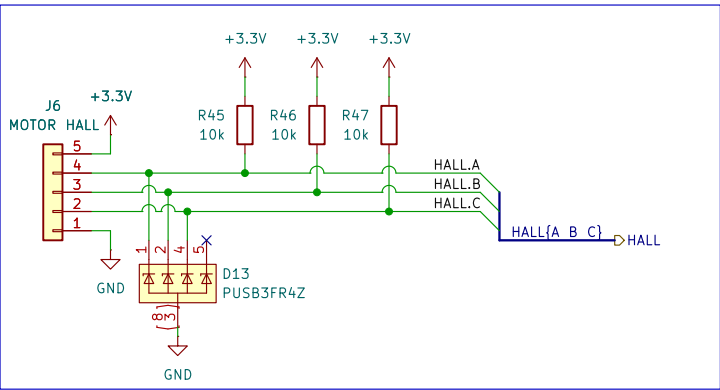
Rev: v1.0

Size: A1	8
KiCad E.D.A. 10.0.3	

Id: 8/10



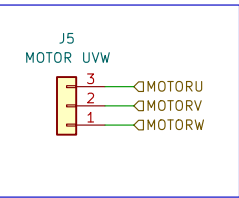
Hall Sensors



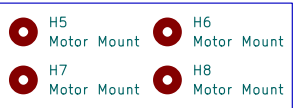
Designed for the Nanotec DF45 65W/50W Motors (no encoder).

The Factory provided connector (STOCK0 MKF 13270-6-0-1010) is not available on DigiKey. This connector must be desoldered and replaced with custom connectors. Use 26 AWG or 24 AWG wire.

Motor Phases



Motor Mounting Holes



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Sheet: /Motor Connections/
File: motor_conn.kicad_sch

Title:

Size: A4

Date: 2026-05-25

Rev: v1.0

KiCad E.D.A. 10.0.3

Id: 10/10